INCH-POUND

MIL-P-43988C
15 June 1993
SUPERSEDING
MIL-P-43988B
29 June 1988

# MILITARY SPECIFICATION

## PAPER, TOILET TISSUE (MRE-PACKET)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a packet of 24 single-ply sheets of toilet tissue for inclusion in Meal Ready To Eat (MRE) packet.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8540

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:17Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 70-82)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3905 - Toilet Tissue, Industrial
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:17Z
Check the source to verify that this is the current version before use.

3.2 Material. The toilet tissue shall be unglazed tissue made from fibrous, cellulosic material which may include reclaimed fibers. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Color. Unless otherwise specified (see 6.2), the color of the toilet tissue shall be white or a uniform shade of dull beige, green, or yellow. Colored tissue shall conform to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.3 Physical requirements. The toilet tissue shall conform to the requirements specified in ASTM D 3905 for type II toilet tissue.

3.4 Assemblage. A packet shall consist of 24 single-ply sheets with approximate dimensions of 4-1/2 by 4-1/2 inches. The sheets shall either be continuous with perforations at approximately 4-1/2 inch intervals or 12 sheet section. The paper shall be folded and wrapped in a sleeve made of unbleached kraft or unbleached sulfite paper. The sleeve shall cover the entire length of the packet within ± 1/8 inch. The sleeve shall be closed by gluing, taping, heat sealing or by means of a self-sticking adhesive applied on opposite sides of both ends of the sleeve to provide a continuous and effective seal. The method of seal shall be such as to ensure that the tissue does not adhere to the sleeve. The method of seal shall not impart an objectionable odor. All edges of the tissue paper visible from the ends of the packet shall be flush with a tolerance of ± 3/16 inch. The maximum dimensions of the final packet shall be not greater than 2-3/4 inches in length, 1-3/8 inches in width, and 5/8 inch in thickness. Alternatively, the sheets shall be enclosed in a sleeve, and the sleeve shall be folded and effectively secured in position by an encircling tape band provided all other requirements for the toilet tissue, sleeve, and finished packet are adhered to and the tape is approximately natural kraft in color.

3.5 Marking. The name and address of the contractor shall be legibly printed or stamped on each packet with water insoluble black ink. The ink shall not bleed through the packet or sleeve or onto the sheets.

3.6 Cleanliness. The toilet tissue shall be clean and free of dirt spots, holes, or tears 1/16 inch or larger, or more than five dirt spots, holes or tears less than 1/16 inch in one sheet; shives or slivers 1/8 inch or more in length, or more than five shives or slivers less than 1/8 inch in length in one sheet; any foreign matter, undigested paper, or unbleached waste. The tissue shall be trimmed with clean, smooth edges.

3.7 Workmanship. The toilet tissue shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:17Z
Check the source to verify that this is the current version before use.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection systems or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and tested as specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for defects listed in table I. The lot size shall be expressed in units of packets. The sample unit shall be one packet. The inspection level shall be



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:17Z
Check the source to verify that this is the current version before use.

I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65 for major defects and 4.0 for total (major and minor combined) defects.

TABLE I. End item visual defects

| Examine | Defect | Classification Major | Classification Minor |
|---|---|---|---|
| Appearance | Formation not uniform | | 201 |
| | Hard or glazed finish | | 202 |
| | Not clean; presence of dirt or spot | 101 | |
| | Holes, tears, shives, or slivers larger than specified or greater in number than specified | | 203 |
| | Presence of foreign matter, undigested paper, or unbleached waste | | 204 |
| | Not trimmed with clean smooth edges | | 205 |
| Color | Other than specified | | 206 |
| | Not uniform | | 207 |
| Packet | Sleeve missing, damaged, or not securely closed | 102 | |
| | Sleeve and tissue adhered to each other | | 208 |
| Marking | Omitted, incorrect, or illegible | | 209 |
| | Ink bleeds onto tissue | | 210 |

4.4.3 End item dimensional examination. The end items shall be examined for the dimensional defects listed below. The lot size shall be expressed in units of packet. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

| Examine | Defect |
|---|---|
| Packet length | More than 2-3/4 inches |
| Packet width | More than 1-3/8 inches |
| Packet thickness | More than 5/8 inch |

4.4.4 Count examination. The end items shall be examined for the number of sheets per packet. The lot size shall be expressed in units of packets. The sample unit shall be one packet. The inspection level shall be S-2 and the lot shall be unacceptable if the minimum number of sheets per packet is less than specified on any of the sample units examined.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:17Z
Check the source to verify that this is the current version before use.

4.4.5 End item testing. The end items shall be tested in accordance with ASTM D 3905. The sample unit shall be three packets. The sample shall consist of five sample units drawn at random from the lot. The lot shall be rejected if the lot average for any characteristic fails to meet the specified requirement. All test reports shall contain the individual values used in calculating the lot averages.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

| Examine | Defect |
|---|---|
| Marking | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |
| Materials | Any component missing, damaged, or not as specified |
| Workmanship | Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling<br>Bulged or distorted container |

5. PACKAGING

5.1 Preservation. Not applicable.

5.2 Packing. Packing shall be commercial.

5.2.1 Commercial packing. The packets shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The toilet tissue packet is for use in the MRE as a component in an accessory packet.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:17Z
Check the source to verify that this is the current version before use.

c. When a first article is required (see 3.1, 4.3 and 6.3).
d. When a specific color or colors are required (see 3.2.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Accessory packet
Paper
Toilet tissue

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 84
DLA - SS

Preparing activity:

Army - GL

(Project 8540-0038)



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:17Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I. RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-P-43988C | 1993 June 15 |

**3. DOCUMENT TITLE**

PAPER, TOILET TISSUE (MRE-PACKET)

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| U.S. Army Natick RD&E Center | (1) Commercial<br>508-651-4532 | (2) AUTOVON/DSN<br>256-4532 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Natick RD&E Center<br>ATTN: SATNC-IRT<br>Natick, MA 01760-5019 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:17Z
Check the source to verify that this is the current version before use.